

GUT Threshold Corrections — The Minimal SU(5) Limit

The threshold coefficients are too small. Minimal SU(5) is disfavored.

Registry: [HOWL-PHYS-29-2026]

Series Path: [HOWL-PHYS-1-2026] → [HOWL-PHYS-13-2026] → [HOWL-PHYS-21-2026] → [HOWL-PHYS-24-2026] → [HOWL-PHYS-25-2026] → [HOWL-PHYS-26-2026] → [HOWL-PHYS-27-2026] → [HOWL-PHYS-28-2026] → [HOWL-PHYS-29-2026]

Date: April 2 2026

Domain: GUT Completion, Threshold Corrections, Null Result

DOI: 10.5281/zenodo.19666420

Status: Complete

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Backed by: phys29_gut_thresholds.py (10/11 checks, 1 FAIL is the abort test firing), phys24_lib.py (21/21 self-test, 148/148 platform test)

Abstract

The two-loop running of the gauge couplings with the Cabibbo Doublet modified betas leaves a residual unification miss $\Delta = -0.40$ (PHYS-24, PHYS-28). This paper computes the GUT threshold corrections in minimal SU(5) that could close this residual. In minimal SU(5), the superheavy spectrum has three components: the X,Y gauge bosons defining M_{GUT} , the colored Higgs triplet $T(3,1,-1/3)$ with beta shift $db_3 = 1/12$, and the Sigma field remnants — an $(8,1,0)$ color octet with $db_3 = 1/2$ and a $(1,3,0)$ weak triplet with $db_2 = 1/3$. The threshold correction to Delta is $\delta_{\Delta} = -(1/12)\ln(M_T/M_X)/(2\pi) - (1/6)\ln(M_{\text{Sigma}}/M_X)/(2\pi)$. The combined coefficient $C_{\text{total}} = -1/4$. All coefficients are exact Fractions from representation theory. To close $\Delta = -0.40$, even the best case (triplet and Sigma at the same mass below M_X) requires $M_X/M = 23,228$ — a mass splitting far beyond natural. The abort test fires: minimal SU(5) is disfavored as the GUT completion. The Cabibbo Doublet representation $(3,2,1/6)$ and its Level 1 arithmetic survive intact — they are independent of the GUT completion. The proton lifetime prediction ($10^{34.5}$ yr) is unchanged.

1. The Two-Loop Residual

The gauge couplings of the Standard Model — the U(1) hypercharge coupling α_1 , the SU(2) weak coupling α_2 , and the SU(3) strong coupling α_3 — run with energy under the renormalization group equations. The inverse couplings $1/\alpha_i$ evolve linearly at one loop and receive corrections at two loops from the 3×3 matrix b_{ij} . If the three gauge forces unify at a single scale M_{GUT} , all three inverse couplings meet at one point: $1/\alpha_1 = 1/\alpha_2 = 1/\alpha_3 = 1/\alpha_{\text{GUT}}$.

The unification miss $\Delta = 1/\alpha_3(M_{\text{GUT}}) - 1/\alpha_{\text{GUT}}$ measures how far α_3 is from the crossing of α_1 and α_2 at M_{GUT} . Negative Δ means α_3 is too strong at the crossing — $1/\alpha_3$ is below the unification value.

The Cabibbo Doublet — a vector-like quark doublet in the (3,2,1/6) representation with modified betas (25/6, -13/6, -20/3) — improves unification at each perturbative order. At one loop with $M_{\text{VL}} = 500$ GeV, $\Delta = -1.17$ (PHYS-24). At two loops with the SM b_{ij} matrix, Δ improves to -0.40, a 66% reduction (PHYS-24). Adding the VL two-loop b_{ij} corrections gives $\Delta = -0.436$, a 63% improvement (PHYS-28).

The remaining $\Delta \approx -0.40$ must be closed by GUT threshold corrections — the effect of superheavy particles having different masses rather than all sitting exactly at M_{GUT} .

(Backed by phys29_gut_thresholds.py Section 1: Delta values from library.)

2. The Minimal SU(5) Heavy Spectrum

In minimal SU(5), the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ emerges when SU(5) breaks at the GUT scale. The breaking is accomplished by a Higgs field in the 24-dimensional adjoint representation (the Sigma field) acquiring a vacuum expectation value. The superheavy spectrum has three components.

The X,Y gauge bosons: 12 gauge degrees of freedom in the (3,2,5/6) + conjugate representation. These acquire mass from the Sigma field VEV and define the GUT scale: $M_X = M_{\text{GUT}}$. They mediate proton decay through dimension-6 operators.

The colored Higgs triplet $T(3,1,-1/3)$: a complex scalar from the fundamental 5 of SU(5). The 5 splits into a weak doublet (which becomes the SM Higgs) and a color triplet (which must be superheavy to avoid rapid proton decay). Its one-loop beta shifts are computed from the scalar Dynkin formulas of PHYS-26: $db_1 = (1/5) \times 3 \times 1 \times (1/9) = 1/15$, $db_2 = 0$ (SU(2) singlet), $db_3 = (1/6) \times 1 \times (1/2) = 1/12$.

The Sigma field remnants: after the 24 adjoint breaks $SU(5) \rightarrow \text{SM}$, 12 of its 24 real components become the longitudinal modes of the X,Y bosons (eaten by the Higgs mechanism). The remaining 12 components form three massive multiplets. The (8,1,0) color octet is a real scalar with 8 components. Its beta shift for SU(3) is $db_3 = (1/6) \times 1 \times S_2(\text{adj } SU(3)) = (1/6) \times 1 \times 3 = 1/2$, where $S_2(\text{adj}) = N = 3$ for the adjoint of SU(3). The (1,3,0) weak triplet is a real scalar with 3 components. Its beta shift for SU(2) is $db_2 =$

$(1/6) \times 1 \times S_2(\text{adj SU}(2)) = (1/6) \times 1 \times 2 = 1/3$, where $S_2(\text{adj}) = N = 2$ for the adjoint of $\text{SU}(2)$. The $(1,1,0)$ singlet has no gauge interactions and contributes nothing to the betas.

(Backed by phys29_gut_thresholds.py S1: $\text{db}_1\text{T} = 1/15$ EXACT, $\text{db}_2\text{T} = 0$ EXACT, $\text{db}_3\text{T} = 1/12$ EXACT.)

3. The Threshold Correction Formula

When the superheavy particles have different masses, the effective running between the lightest and heaviest superheavy scale uses different betas in different energy intervals, introducing a threshold correction to the unification prediction.

The correction to Delta has the form:

$$\text{delta_Delta} = C_T \times \ln(M_T/M_X)/(2 \pi) + C_Sigma \times \ln(M_Sigma/M_X)/(2 \pi)$$

where M_X is the X boson mass (defining M_GUT), M_T is the colored triplet mass, M_Sigma is the Sigma remnant mass, and C_T and C_Sigma are the effective threshold coefficients.

The triplet coefficient: $C_T = -\text{db}_3\text{T} = -1/12$. The triplet shifts only the α_3 running ($\text{db}_2\text{T} = 0$ means the $\alpha_1 = \alpha_2$ crossing is unaffected). If $M_T < M_X$, the triplet is active for a longer range, providing more correction to $1/\alpha_3$ at M_GUT .

The Sigma coefficient: $C_Sigma = \text{db}_2\text{Sigma} - \text{db}_3\text{Sigma} = 1/3 - 1/2 = -1/6$. This accounts for two effects: the $(8,1,0)$ octet shifts α_3 running by $\text{db}_3 = 1/2$, and the $(1,3,0)$ weak triplet shifts α_2 running by $\text{db}_2 = 1/3$, which shifts the $\alpha_1 = \alpha_2$ crossing point and hence $1/\alpha_GUT$. The net effect on Delta is the difference.

The combined coefficient: $C_total = C_T + C_Sigma = -1/12 - 1/6 = -3/12 = -1/4$. If both the triplet and the Sigma remnants are at the same mass M below M_X :

$$\text{delta_Delta} = -(1/4) \times \ln(M/M_X)/(2 \pi)$$

All three coefficients $(-1/12, -1/6, -1/4)$ are exact Fractions from the representation theory of $\text{SU}(5)$. They are Level 1.

(Backed by phys29_gut_thresholds.py S2: $C_T = -1/12$ EXACT, $C_Sigma = -1/6$ EXACT, $C_total = -1/4$ EXACT.)

4. Closing the Residual — Three Cases

To close the two-loop residual $\Delta = -0.40$, the threshold correction must provide $\text{delta_Delta} = +0.40$. Since all threshold coefficients are negative, this requires $\ln(M/M_X) < 0$, meaning the heavy scalars must be LIGHTER than the X bosons.

Case 1: Triplet only (Sigma at M_X). $C = -1/12$. Required: $-(1/12) \times \ln(M_T/M_X)/(2\pi) = 0.40$. Solving: $\ln(M_T/M_X) = -0.40 \times 24\pi = -30.2$. The ratio $M_X/M_T = e^{30.2} = 1.25 \times 10^{13}$.

Case 2: Sigma only (triplet at M_X). $C = -1/6$. Required: $\ln(M_{\text{Sigma}}/M_X) = -0.40 \times 12\pi = -15.1$. The ratio $M_X/M_{\text{Sigma}} = 3.5 \times 10^6$.

Case 3: Both at same mass. $C = -1/4$. Required: $\ln(M/M_X) = -0.40 \times 8\pi = -10.1$. The ratio $M_X/M = 23,228$.

All three cases require mass ratios far exceeding the naturalness threshold of 10 and far exceeding the fine-tuning threshold of 100. Even the best case (both at the same mass, maximizing the combined coefficient) requires a factor of 23,000 between the X boson mass and the scalar masses.

(Backed by phys29_gut_thresholds.py Section 3: all three cases computed.)

5. Why the Coefficients Are Small

The fundamental limitation is structural. The threshold coefficients $C = -1/12, -1/6, -1/4$ are small fractions. They are small because they involve the Dynkin indices of fundamental and adjoint representations of $SU(2)$ and $SU(3)$ — numbers of order 1/2 to 3 — divided by the scalar counting factors (1/6 for real scalars, 1/12 for complex scalars in the effective coefficient). The formula $\text{delta_Delta} = C \times \ln(M/M_X)/(2\pi)$ divides by an additional factor of $2\pi \approx 6.28$, giving an effective coefficient $C/(2\pi) \approx 0.01$ to 0.04.

To produce $\text{delta_Delta} = 0.40$ from an effective coefficient of 0.04, the logarithm must be ~ 10 , corresponding to a mass ratio of $\sim e^{10} \approx 22,000$.

This is a property of minimal $SU(5)$ specifically. A GUT completion with larger representations (higher Dynkin indices), more heavy particles (cumulative effect from multiple thresholds), or intermediate-scale breaking (additional running segments) could have larger effective coefficients. The threshold correction scan in the script confirms: even $M_X/M = 500$ provides only $\text{delta_Delta} = 0.082$ against the required 0.40.

6. The Null Result

The abort test stated in PHYS-25 was: “If exact unification requires $M_T/M_X > 100$, the minimal $SU(5)$ completion is disfavored.” The best case gives $M_X/M = 23,228$. The abort test fires.

Minimal $SU(5)$ is disfavored as the GUT completion for the Cabibbo Doublet framework.

What survives:

The Cabibbo Doublet (3,2,1/6) and all its Level 1 arithmetic. The beta shifts (1/15, 1, 1/3), the gap ratio 38/27, the modified betas (25/6, -13/6, -20/3), and all integers derived from them (13, 20, 22, etc.) are independent of the GUT completion. They are determined by the representation under $SU(3) \times SU(2) \times U(1)$, not by the embedding into a larger group.

The two-loop improvement (66%). The SM b_{ij} matrix is measured and model-independent. The VL b_{ij} contribution depends only on the CD representation, not on the GUT.

The proton lifetime prediction. The X boson mass $M_X = M_{GUT} \approx 10^{15.4}$ GeV is set by the coupling crossing, which depends on the CD one-loop betas (Level 1) and the measured couplings (Level 2). It is unchanged by the threshold finding. The proton decay channel $p \rightarrow e^+ \pi^0$ mediated by X boson exchange gives $\tau \sim M_X^4/\alpha_{GUT}^2 \sim 10^{34.5}$ yr, above the Super-K bound of 2.4×10^{34} yr and within the Hyper-K 20-year sensitivity of $\sim 10^{35}$ yr.

What is disfavored:

The specific claim that minimal $SU(5)$ — with only the 5 (Higgs) and 24 (Sigma) representations breaking $SU(5) \rightarrow SM$ — can achieve exact unification with natural mass splittings. The residual $\Delta = -0.40$ requires threshold corrections beyond what these representations provide.

(Backed by `phys29_gut_thresholds.py` S4: abort test fires with $M_X/M = 23,228$.)

7. Alternative Pathways

Three classes of alternatives to minimal $SU(5)$ could provide larger threshold corrections:

SO(10) or Pati-Salam intermediate stages: in SO(10), the breaking can proceed through $SU(5) \times U(1)$ or $SU(4) \times SU(2) \times SU(2)$ intermediate groups. The additional heavy particles at the intermediate scale — leptoquarks, right-handed W bosons, additional Higgs multiplets — provide threshold coefficients from higher representations with larger Dynkin indices. The intermediate-scale running also changes the effective one-loop betas between $M_{intermediate}$ and M_{GUT} , potentially reducing the residual Δ before thresholds are needed.

Extended Higgs sector: adding heavy scalar representations beyond the minimal 5 + 24 of $SU(5)$ increases the cumulative threshold coefficient. Each additional representation contributes its own C term. A handful of carefully chosen representations could accumulate enough correction with natural mass splittings.

Higher perturbative orders: the three-loop beta function provides additional correction comparable in magnitude to the two-loop VL effect ($\sim 5\%$ of one-loop Δ , per PHYS-28). If the three-loop correction goes in the right direction, it reduces the residual that the thresholds must close, easing the naturalness requirement.

This paper does not pursue these alternatives. They are staged for future investigation. The finding is that minimal SU(5) is insufficient. The alternatives are stated, not computed.

8. What This Paper Does Not Claim

This paper does not claim the Cabibbo Doublet is wrong. The CD is Level 1 arithmetic — its beta shifts, gap ratio, and derived integers are representation theory, not GUT model-building. The null result is about the GUT completion, not about the CD.

This paper does not claim gauge coupling unification is impossible. It claims exact unification in MINIMAL SU(5) requires unnatural mass splittings. Non-minimal completions with more heavy particles or intermediate-scale breaking may achieve natural unification.

This paper does not claim the threshold formulas are exact. The perturbative threshold correction is a leading-order result. Higher-order threshold corrections, the running of heavy particle masses, and Yukawa coupling effects introduce corrections at the few-percent level.

This paper does not claim the two-loop residual is precisely -0.40 . This value comes from the PHYS-24 higher-order integrator. The PHYS-28 Euler integration gives -0.436 . A more precise residual would change the required mass ratio but not the structural conclusion — the threshold coefficients are too small for natural splittings regardless of whether the residual is -0.35 or -0.45 .

9. What This Paper Seeds

The null result redirects the unification completion program:

PHYS-30 (α_s prediction) proceeds unchanged. The α_s prediction uses the two-loop running with the threshold uncertainty as a systematic, not as a tuned parameter.

A future paper on SO(10) intermediate-scale breaking would compute the threshold coefficients for the extended heavy spectrum and test whether they achieve natural unification. This is the most promising alternative.

A future paper on the extended Higgs sector would enumerate heavy scalar additions to minimal SU(5) and compute cumulative thresholds.

The exact Fraction threshold coefficients ($C_T = -1/12$, $C_{\text{Sigma}} = -1/6$, $C_{\text{total}} = -1/4$) are published for any future computation requiring them. The Sigma remnant beta shifts ($db_3 = 1/2$ for the octet, $db_2 = 1/3$ for the weak triplet) are new results not in the PHYS-24 library and should be added to the platform for future use.

PHYS-29: GUT Threshold Corrections. The minimal SU(5) limit. $C_{total} = -1/4$. $M_X/M = 23,228$. Abort test fires. 10/11 checks (1 FAIL is the abort test). Published April 2, 2026. This paper is never edited after publication.

Appendix A: The Minimal SU(5) Heavy Spectrum

Particle	Representation	Type	DOF	db ₁	db ₂	db ₃	Role
X,Y bosons	(3,2,5/6) + conj	Vector	12	—	—	—	Define M GUT
Triplet T	(3,1,-1/3)	Complex scalar	6	1/15	0	1/12	Higgs sector
Sigma (8,1,0)	Adjoint of SU(3)	Real scalar	8	0	0	1/2	Sigma remnant
Sigma (1,3,0)	Adjoint of SU(2)	Real scalar	3	0	1/3	0	Sigma remnant
Sigma (1,1,0)	Singlet	Real scalar	1	0	0	0	Sigma remnant

Appendix B: The Threshold Coefficients

Coefficient	Formula	Value	Origin
C T	-db ₃ T	-1/12	Triplet shifts α_3 only (db ₂ T = 0)
C Sigma	db ₂ Sigma - db ₃ Sigma	1/3 - 1/2 = -1/6	Octet shifts α_3 , weak triplet shifts crossing
C total	C T + C Sigma	-1/12 - 1/6 = -1/4	Combined
C T/(2 π)	effective coefficient	-0.013	Per unit $\ln(M T/M X)$
C Sigma/(2 π)	effective coefficient	-0.027	Per unit $\ln(M \text{ Sigma}/M X)$
C total/(2 π)	effective coefficient	-0.040	Per unit $\ln(M/M X)$

Appendix C: The Three Cases — Closing Delta = -0.40

Case	Active particle	C	Required ln	Required ratio M X/M	Natural?
1: Triplet only	T(3,1,-1/3)	-1/12	-30.2	1.25×10^{13}	No
2: Sigma only	(8,1,0) + (1,3,0)	-1/6	-15.1	3.5×10^6	No
3: Both at same mass	All	-1/4	-10.1	23,228	No

Naturalness threshold: $M_X/M < 10$. Fine-tuning threshold: $M_X/M > 100$. All three cases exceed the fine-tuning threshold.

Appendix D: Threshold Correction Scan

M X/M T	ln(M T/M X)	delta Delta	Delta total
1	0.000	0.000	-1.172
2	-0.693	0.009	-1.163
3	-1.099	0.015	-1.157
5	-1.609	0.021	-1.150
10	-2.303	0.031	-1.141
20	-2.996	0.040	-1.132
50	-3.912	0.052	-1.120
100	-4.605	0.061	-1.111
200	-5.298	0.070	-1.102
500	-6.215	0.082	-1.089

Even $M_X/M_T = 500$ shifts Delta by only 0.082, from -1.172 to -1.089. The two-loop correction (shifting Delta from -1.17 to -0.40) does far more work than any natural threshold correction.

Appendix E: What Survives and What Is Disfavored

Item	Status	Reason
CD representation (3,2,1/6)	Survives	Level 1 arithmetic, independent of GUT
Beta shifts (1/15, 1, 1/3)	Survives	Dynkin formulas, independent of GUT

Item	Status	Reason
Gap ratio 38/27	Survives	Exact Fraction, independent of GUT
Integers 13, 20, 22	Survive	From modified betas, independent of GUT
Two-loop improvement (66%)	Survives	SM b_{ij} is model-independent
VL two-loop improvement (+4.6%)	Survives	From CD representation theory
$M_{\text{GUT}} \approx 10^{15.4} \text{ GeV}$	Survives	From coupling crossing
$\tau_{\text{proton}} \sim 10^{34.5} \text{ yr}$	Survives	From M_X^4 scaling
Cosmological formulas	Survive	From integers 13, 20, independent of GUT
$\sin^2 \theta_W \approx 3/13$	Survives	From running, independent of GUT completion
Minimal SU(5) completion	Disfavored	$M_X/M = 23,228$ required
Natural GUT thresholds in min SU(5)	Disfavored	Coefficients too small

Appendix F: Verification Summary

Check	Description	Status
S1	Triplet $db_1 = 1/15$	PASS (EXACT)
S1	Triplet $db_2 = 0$	PASS (EXACT)
S1	Triplet $db_3 = 1/12$	PASS (EXACT)
S2	$C_T = -1/12$	PASS (EXACT)
S2	$C_{\text{Sigma}} = -1/6$	PASS (EXACT)
S2	$C_{\text{total}} = -1/4$	PASS (EXACT)
S3	Triplet-only needs $M_T < M_X$	PASS
S3	Both-at-same-mass needs $M < M_X$	PASS
S4	Abort test: $M_X/M < 100$	FAIL ($M_X/M = 23,228$)
S5	$M_{\text{GUT}} > 10^{15}$	PASS
S5	Proton lifetime above Super-K	PASS
Total		10 PASS, 1 FAIL

The single FAIL is the abort test firing — a physics result, not a script bug. The abort test was designed to fire when minimal SU(5) requires unnatural mass splittings. It fires correctly.

Supporting appendices A through F for PHYS-29. Every threshold coefficient is exact. The null result is documented with full numerical support. The Cabibbo Doublet and all Level 1 arithmetic survive. Grand total across all scripts: 466/467 (1 FAIL is the PHYS-29 abort test).

[[HOWL-PHYS-29-2026](#)] GUT Threshold Corrections — The Minimal SU(5) Limit. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-29-2026>

[[HOWL-PHYS-1-2026](#)] The Inertial Vortex. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-1-2026>

[[HOWL-PHYS-13-2026](#)] Gauge Coupling Unification and Minimal BSM Content. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-13-2026>

[[HOWL-PHYS-21-2026](#)] The Level 1 / Level 2 Boundary — From Observed Structure to Unobserved Particles. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-21-2026>

[[HOWL-PHYS-24-2026](#)] The Session 3 Operational Lexicon. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS-24-2026>

[[HOWL-PHYS-25-2026](#)] The Session 4 Operational Map — From Gauge Integers to Cosmological Predictions. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-25-2026>

[[HOWL-PHYS-26-2026](#)] The Integer Traceability Chain. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS-26-2026>

[[HOWL-PHYS-27-2026](#)] $\sin^2\theta_W$ from $3/8$ — The Weak Mixing Angle as a Running Prediction. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-27-2026>

[[HOWL-PHYS-28-2026](#)] The VL Two-Loop Beta Matrix — Fermion Corrections to Gauge Coupling Unification. Github: <https://github.com/ghowland/papers/tree/main/papers/PHYS/HOWL-PHYS-28-2026>